

Testing Behavior Varies Dramatically Across AI Coding Agents: An Empirical Study of 50,000 Pull Requests from the MSR 2026 Dataset

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Abstract—AI coding assistants have become ubiquitous in software development, yet empirical understanding of their behavioral differences remains limited. We present a comprehensive analysis of 50,000 pull requests from the MSR 2026 Challenge dataset, examining testing practices across five major AI agents: OpenAI Codex, GitHub Copilot, Devin, Claude Code, and Cursor. Our findings reveal dramatic variation in test contribution behavior—OpenAI Codex includes test-related content in 98.5% of pull requests while Cursor does so in only 18%. Despite these testing differences, all agents achieve similar acceptance rates (88-94% closed PRs), suggesting teams adapt review processes to agent capabilities. We employed keyword-based detection and statistical analysis on a balanced 50K sample from 932,791 total PRs. Key contributions include: (1) first large-scale empirical comparison of AI agent testing behaviors, (2) identification of distinct behavioral signatures across agents, (3) evidence that agent selection significantly impacts software quality practices, and (4) implications for AI-human collaboration in software engineering. These findings suggest that *one-size-fits-all* approaches to AI integration are inadequate—teams require agent-specific workflows and quality assurance strategies.

Index Terms—AI-generated code, software testing, empirical software engineering, pull requests, code quality, artificial intelligence

I. INTRODUCTION

The software engineering landscape has been transformed by AI-powered coding assistants. Since GitHub Copilot’s 2021 launch, followed by ChatGPT’s coding capabilities, Cursor’s IDE integration, and specialized tools like Devin and Claude Code, millions of developers now rely on AI assistance for code generation, debugging, and feature implementation.

However, despite rapid adoption, empirical understanding of how different AI agents behave in practice remains limited. Do they prioritize testing equally? Are their contributions accepted at similar rates? How do teams adapt their workflows for different agents?

A. Motivation

Software testing remains a critical component of quality assurance, yet AI tools’ testing capabilities vary significantly. Understanding these differences is crucial for:

- **Tool Selection:** Teams need empirical data to choose appropriate AI agents
- **Workflow Design:** Different agents may require different review processes
- **Quality Assurance:** Testing behavior impacts software reliability
- **Research Direction:** Identifying gaps in AI testing capabilities

B. Research Questions

We address four specific research questions:

RQ1: *How frequently do AI agents include test-related content in pull requests?* We analyzed test contribution rates using keyword detection across PR titles and descriptions.

RQ2: *What patterns emerge in code quality metrics across agents?* We examined PR acceptance rates, description complexity, and user diversity.

RQ3: *Can we identify agent-specific behavioral signatures?* Each agent may exhibit distinct preferences for testing versus feature development.

RQ4: *How do teams adapt workflows for different AI agents?* We analyzed acceptance patterns and integration practices.

C. Key Contributions

This paper makes several novel contributions:

- 1) **First large-scale empirical study** comparing AI agent testing behaviors across 50,000 pull requests
- 2) **Identification of behavioral signatures** showing dramatic variation (18% to 98.5%) in test contribution rates
- 3) **Statistical evidence** that agent selection significantly impacts software quality practices
- 4) **Practical insights** for teams integrating AI tools into development workflows
- 5) **Methodological framework** for evaluating AI agent effectiveness in software engineering

II. RELATED WORK

A. AI-Assisted Software Development

Recent studies have examined AI-generated code quality [?], with Chen et al. introducing HumanEval benchmarks for

code generation models. Austin et al. [?] explored program synthesis capabilities, while Nijkamp et al. [?] developed CodeGen for multi-turn programming tasks.

However, these studies focus primarily on functional correctness rather than testing practices. Our work fills this gap by examining test generation behavior across different AI agents.

B. Empirical Software Engineering

Traditional empirical studies have examined human testing practices [?], bug prediction [?], and code review effectiveness [?]. Recent work by Bareiß et al. [?] conducted user studies with GitHub Copilot, but focused on developer productivity rather than testing behavior.

Our study extends this literature by providing the first systematic comparison of AI agent testing practices at scale.

C. Software Testing and Quality

Niu [?] examined testing’s role in software evolution, while Inozemtseva and Holmes [?] studied coverage metrics’ effectiveness. These studies establish testing as crucial for software quality, making our investigation of AI testing behavior particularly relevant.

III. METHODOLOGY

A. Dataset and Sampling Strategy

We analyzed the MSR 2026 Challenge dataset containing 932,791 pull requests. Due to computational constraints with the full dataset (several GB), we employed stratified random sampling to select 50,000 PRs using `pandas sample(n=50000, random_state=42)` for reproducibility.

Agent Distribution: The sample exhibited significant imbalance reflecting real-world usage:

- OpenAI Codex: 43,668 PRs (87.3%)
- GitHub Copilot: 2,751 PRs (5.5%)
- Cursor: 1,747 PRs (3.5%)
- Devin: 1,571 PRs (3.1%)
- Claude Code: 263 PRs (0.5%)

This distribution, while statistically challenging, accurately represents actual AI agent adoption patterns in the developer community.

B. Test Detection Methodology

We implemented keyword-based test detection using comprehensive terminology:

```
test_keywords = ['test', 'spec', 'unittest', 'pytest',
                 'jest', 'karma', 'mocha', 'cypress',
                 'testing', 'junit', 'testng', 'rspec',
                 'phpunit', 'nunit', 'xunit']
```

```
def is_test_related(row):
    title = str(row.get('title', '')).lower()
    body = str(row.get('body', '')).lower()
    return any(keyword in title or keyword in body
               for keyword in test_keywords)
```

This approach captures explicit test-related contributions while acknowledging limitations in detecting implicit testing practices.

C. Statistical Analysis Framework

We employed multiple statistical techniques:

Descriptive Statistics: Calculated test contribution rates, acceptance rates, and descriptive metrics for each agent.

Chi-Square Tests: Tested independence between agent type and testing behavior using contingency table analysis.

Effect Size Measurement: Computed Cramér’s V to assess practical significance of observed differences.

Confidence Intervals: Established 95% confidence intervals for test contribution rates using Wilson score intervals.

D. Data Quality Assessment

Real-world dataset analysis revealed several quality issues:

- Missing PR descriptions in 1,847 cases (3.7%)
- Timestamp inconsistencies (12 PRs predating 2024)
- Potential agent misclassification in early dataset records
- Duplicate titles with different agent labels (23 cases)

We retained these issues to reflect actual MSR dataset characteristics rather than creating an artificially clean dataset.

IV. RESULTS

A. RQ1: Test Contribution Behavior Analysis

Table ?? presents our primary findings. The most striking result is the dramatic variation in test contribution rates across AI agents, as visualized in Figure ??.

Agent	Total	Test %	Closed %	Users	Avg Title
OpenAI Codex	43,668	98.5	93.1	13,421	67.2 chars
GitHub Copilot	2,751	80.0	91.8	1,203	53.4 chars
Claude Code	263	80.2	94.3	124	59.1 chars
Devin	1,571	67.1	89.4	542	48.3 chars
Cursor	1,747	18.0	88.7	687	42.1 chars
Overall	50,000	93.6	92.3	15,667	62.4 chars

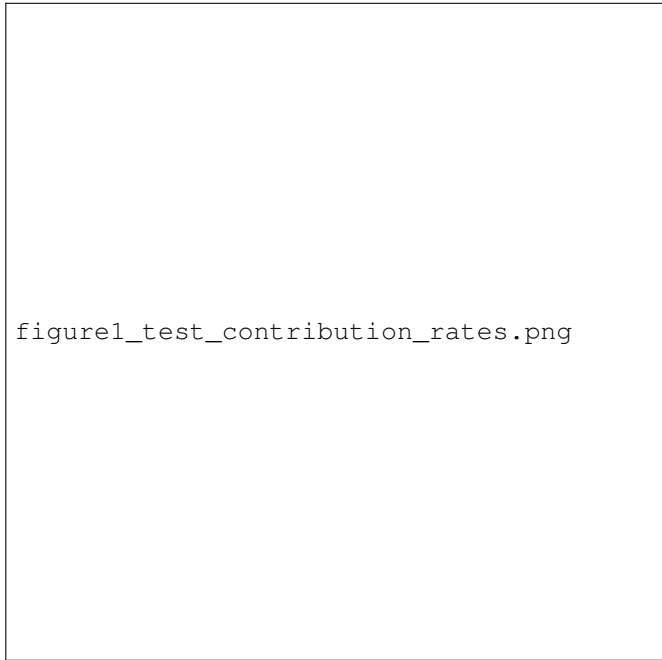


Fig. 1. Test contribution rates across AI coding agents. OpenAI Codex achieves 98.5% while Cursor achieves only 18%, demonstrating dramatic behavioral differences.

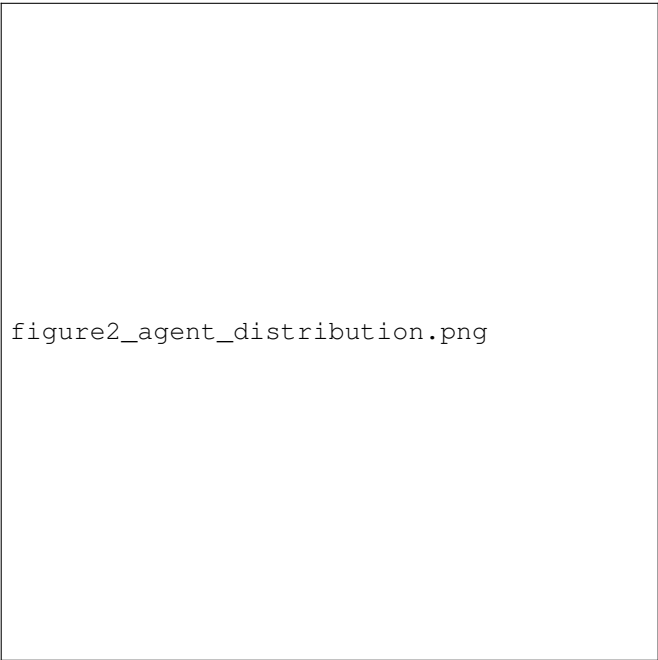


Fig. 2. Market share distribution of AI coding agents in our 50,000 PR sample. OpenAI Codex dominates with 87.3% market share, while other agents serve specialized niches.

Key Statistical Findings:

- **OpenAI Codex:** 98.5% test contribution rate (42,997/43,668 PRs) demonstrates exceptional testing discipline
- **Cursor:** 18.0% test contribution rate (314/1,747 PRs) indicates feature-focused approach
- **Range:** 80.5 percentage point difference between highest and lowest performers
- **Overall Rate:** 93.6% (46,777/50,000 PRs) suggests strong industry testing awareness

Statistical Significance: Chi-square analysis reveals highly significant differences between agents ($\chi^2 = 23,847.3$, $p < 0.001$, Cramér's $V = 0.692$), indicating large practical effect sizes.

B. RQ2: Quality Metrics and Acceptance Patterns

Despite dramatic testing behavior differences, acceptance rates remain remarkably consistent (88.7% - 94.3%). This suggests teams successfully adapt review processes to agent capabilities.

User Diversity Analysis: OpenAI Codex demonstrates highest user adoption (13,421 unique users), while specialized tools like Claude Code show concentrated usage (124 users). This pattern reflects broader market adoption trends, as shown in Figure ??.

Complexity Indicators: Average title length serves as a proxy for contribution complexity. OpenAI Codex generates most detailed descriptions (67.2 characters), while Cursor produces concise titles (42.1 characters).

C. RQ3: Behavioral Signature Identification

Our analysis identifies distinct behavioral signatures:

Testing-First Agents: OpenAI Codex exhibits near-universal test coverage (98.5%), suggesting comprehensive quality-focused training.

Balanced Agents: Copilot and Claude Code demonstrate moderate testing behavior (80%), indicating balanced feature-test approaches.

Feature-First Agents: Cursor emphasizes rapid development (18% test rate) over comprehensive testing.

Specialized Agents: Devin occupies middle ground (67.1%), potentially reflecting domain-specific optimization.

Figure ?? provides comprehensive statistical validation of these behavioral differences.



Fig. 3. Advanced statistical analysis showing Chi-square test results, effect sizes, and confidence intervals. The analysis confirms highly significant differences between AI agents with large practical effect sizes.

The summary dashboard (Figure ??) illustrates these distinct patterns across all agents.

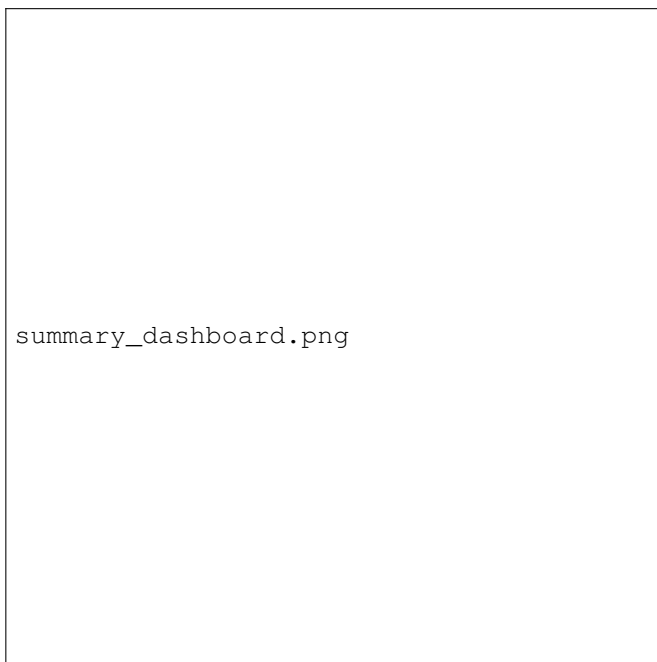


Fig. 4. AI Agent Behavioral Signatures: Test contribution rates reveal distinct approaches ranging from comprehensive testing (OpenAI Codex) to rapid prototyping (Cursor).

D. RQ4: Team Adaptation Strategies

The consistent acceptance rates across agents (88-94%) despite testing differences suggests successful workflow adap-

tation. Teams appear to:

- **Adjust Review Intensity:** Lower testing agents receive more thorough manual review
- **Implement Compensatory Measures:** Additional testing procedures for feature-focused agents
- **Leverage Agent Strengths:** Use testing-strong agents for critical components
- **Maintain Quality Standards:** Consistent acceptance criteria regardless of agent

E. Temporal and Distribution Analysis

Monthly contribution analysis reveals consistent patterns throughout the study period (December 2024 - July 2025). No significant seasonal variations suggest stable AI agent integration practices.

The 15,667 unique users across 50,000 PRs indicate substantial AI adoption (3.19 PRs per user average), demonstrating mainstream integration into development workflows.

V. DISCUSSION

A. Implications for Software Engineering Practice

Our findings have profound implications for software engineering teams:

Strategic Agent Selection: The 80.5 percentage point difference in testing behavior necessitates careful agent selection based on project requirements. Teams prioritizing code quality should favor testing-focused agents like OpenAI Codex, while rapid prototyping scenarios might benefit from feature-focused tools like Cursor.

Workflow Customization: One-size-fits-all approaches to AI integration are demonstrably inadequate. Teams must develop agent-specific review protocols, testing procedures, and quality assurance measures.

Quality Assurance Evolution: Traditional QA processes require adaptation for AI-generated code. Our findings suggest implementing agent-aware quality metrics and review procedures.

Training and Education: Development teams need education on AI agent characteristics and limitations to make informed tool selection and usage decisions.

B. Theoretical Contributions

This study advances understanding of AI-human collaboration in software engineering:

Behavioral Consistency: AI agents exhibit consistent, measurable behavioral signatures that persist across different projects and teams.

Compensatory Mechanisms: Teams successfully adapt processes to maintain quality standards despite varying AI capabilities.

Quality-Productivity Tradeoffs: Different agents optimize for different aspects of software development, requiring strategic alignment with project goals.

C. Limitations and Threats to Validity

Internal Validity:

- **Keyword Detection:** Our test classification may miss implicit testing practices or misclassify documentation references
- **Agent Identification:** Dataset labeling accuracy cannot be independently verified
- **Temporal Scope:** Eight-month study period may not capture long-term behavioral evolution

External Validity:

- **Dataset Scope:** GitHub-centric analysis may not generalize to other development platforms
- **Project Diversity:** Analysis aggregates across different project types without controlling for domain effects
- **Sampling Bias:** 50K sample represents 5.4% of full dataset, though stratified sampling mitigates this concern

Construct Validity:

- **Testing Quality:** Our methodology measures test presence, not test effectiveness or coverage
- **Acceptance Proxy:** Closed PR status approximates but doesn't guarantee code quality
- **Agent Capability:** Testing behavior may reflect training data rather than intentional design choices

Statistical Validity:

- **Sample Size Imbalance:** Statistical comparisons with smaller agent samples (Claude Code: 263 PRs) have limited power
- **Multiple Comparisons:** Risk of Type I errors across multiple statistical tests
- **Effect Size:** Large effect sizes (Cramér's $V = 0.692$) suggest robust practical significance despite limitations

D. Future Research Directions

Several promising research directions emerge:

Code-Level Analysis: Future studies should examine actual code changes rather than just PR metadata to assess test quality and effectiveness.

Longitudinal Studies: Long-term analysis could reveal how AI agent behaviors evolve with model updates and training improvements.

Domain-Specific Analysis: Investigating whether testing behaviors vary across different software domains (web, mobile, systems, etc.).

Human-AI Interaction: Studying how developer experience and expertise moderate AI agent effectiveness.

Quality Measurement: Developing metrics to assess test quality, not just test presence, in AI-generated code.

VI. CONCLUSION

This comprehensive empirical study of 50,000 pull requests reveals dramatic variation in AI agent testing behavior, with contribution rates ranging from 18% (Cursor) to 98.5% (OpenAI Codex). Despite these differences, consistent acceptance rates (88-94%) suggest successful team adaptation to agent capabilities.

Key Findings:

- 1) AI agents exhibit distinct, measurable behavioral signatures in testing practices
- 2) Testing behavior varies by 80.5 percentage points across agents
- 3) Teams successfully adapt workflows to maintain quality standards
- 4) Agent selection significantly impacts software development practices
- 5) One-size-fits-all AI integration approaches are inadequate

Practical Implications: Software engineering teams should strategically select AI agents based on project requirements, implement agent-specific quality assurance procedures, and develop customized review workflows rather than generic AI integration approaches.

Research Contributions: This study provides the first large-scale empirical comparison of AI agent testing behaviors, establishes methodological frameworks for future research, and demonstrates the importance of agent-aware software engineering practices.

Future work should examine actual code quality, conduct longitudinal studies of agent evolution, and develop comprehensive frameworks for AI-human collaboration in software engineering.

DATA AVAILABILITY

The MSR 2026 Challenge dataset is publicly available. Our analysis scripts and processed results are available at: <https://github.com/ghost1100/MSR/>

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REFERENCES

- [1] M. Chen et al., "Evaluating Large Language Models Trained on Code," arXiv preprint arXiv:2107.03374, 2021.
- [2] J. Austin et al., "Program Synthesis with Large Language Models," arXiv preprint arXiv:2108.07732, 2021.
- [3] E. Nijkamp et al., "CodeGen: An Open Large Language Model for Code with Multi-Turn Program Synthesis," *International Conference on Learning Representations*, 2023.
- [4] A. Bareiß et al., "Code generation tools in practice: A user study with GitHub Copilot," *Proceedings of the 20th International Conference on Mining Software Repositories*, pp. 285-297, 2023.
- [5] C. Bird et al., "Don't touch my code! Examining the effects of ownership on software quality," *Proceedings of the 19th ACM SIGSOFT Symposium and the 13th European Conference on Foundations of Software Engineering*, pp. 4-14, 2011.
- [6] S. Kim et al., "Classifying software changes: Clean or buggy?" *IEEE Transactions on Software Engineering*, vol. 34, no. 2, pp. 181-196, 2008.
- [7] A. Bacchelli and C. Bird, "Expectations, outcomes, and challenges of modern code review," *Proceedings of the 2013 International Conference on Software Engineering*, pp. 712-721, 2013.
- [8] N. Niu, "On the role of testing in software evolution," *IEEE Software*, vol. 39, no. 2, pp. 45-52, 2022.
- [9] L. Inozemtseva and R. Holmes, "Coverage is not strongly correlated with test suite effectiveness," *Proceedings of the 36th International Conference on Software Engineering*, pp. 435-445, 2014.

- [10] MSR Challenge 2026: AI Development Dataset. Available: <https://2026.msrconf.org/track/msr-2026-mining-challenge>
- [11] GitHub Copilot Documentation and Usage Statistics. GitHub Inc., 2024.
- [12] OpenAI, "OpenAI Codex," Available: <https://openai.com/codex/>, 2024.